



Unit 7: Firms in Competitive Markets

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Objectives

In this unit, look for the answers to these questions:

- What is a perfectly competitive market?
- What is marginal revenue? How is it related to total and average revenue?
- How does a competitive firm determine the quantity that maximizes profits?
- When might a competitive firm shut down in the short run? Exit the market in the long run?
- What does the market supply curve look like in the short run? In the long run?

2



Introduction: A Scenario

- Three years after graduating, you run your own business.
- You have to decide how much to produce, what price to charge, how many workers to hire, etc.
- What factors should affect these decisions?
 - Your costs (studied in preceding chapter)
 - How much competition you face
- We begin by studying the behavior of firms in perfectly competitive markets.

3



Characteristics of Perfect Competition

1. Many buyers and many sellers
 2. The goods offered for sale are largely the same.
 3. Firms can freely enter or exit the market.
- Because of 1 & 2, each buyer and seller is a “**price taker**” – takes the price as given.

4



The Revenue of a Competitive Firm

- Total revenue (TR)

$$TR = P \times Q$$

- **Average revenue (AR)**

$$AR = \frac{TR}{Q} = P$$

- **Marginal Revenue (MR):**

The change in TR from selling one more unit.

$$MR = \frac{\Delta TR}{\Delta Q}$$

5



ACTIVE LEARNING 1: Exercise

Fill in the empty spaces of the table.

Q	P	TR	AR	MR
0	\$10		n.a.	
1	\$10		\$10	
2	\$10			
3	\$10			
4	\$10	\$40		
5	\$10	\$50		

6



$MR = P$ for a Competitive Firm

- A competitive firm can keep increasing its output without affecting the market price.
- So, each one-unit increase in Q causes revenue to rise by P , i.e., $MR = P$.

$MR = P$ is only true for firms in competitive markets.

7



Profit Maximization

- What Q maximizes the firm's profit?
- To find the answer,
"Think at the margin."
If increase Q by one unit,
revenue rises by MR ,
cost rises by MC .
- If $MR > MC$, then increase Q to raise profit.
- If $MR < MC$, then reduce Q to raise profit.



8



Profit Maximization

(continued from earlier exercise)

Q	TR	TC	Profit	MR	MC	$\Delta\text{Profit} = MR - MC$
0	\$0	\$5	-\$5			
1	10	9	1	\$10	\$4	\$6
2	20	15	5	10	6	4
3	30	23	7	10	8	2
4	40	33	7	10	10	0
5	50	45	5	10	12	-2

At any Q with $MR > MC$, increasing Q raises profit.

At any Q with $MR < MC$, reducing Q raises profit.

9



MC and the Firm's Supply Decision

Rule: $MR = MC$ at the profit-maximizing Q .

At Q_a , $MC < MR$.

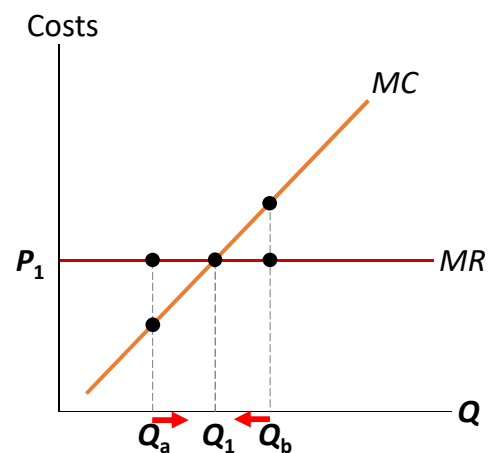
So, increase Q to raise profit.

At Q_b , $MC > MR$.

So, reduce Q to raise profit.

At Q_1 , $MC = MR$.

Changing Q would lower profit.



10



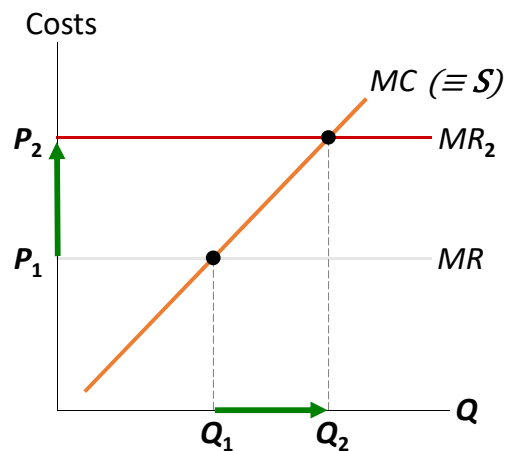
MC and the Firm's Supply Decision

If price rises to P_2 ,
then the profit-maximizing
quantity rises to Q_2 .

The MC curve determines the
firm's Q at any price.

Hence,

the MC curve is the firm's
supply curve.



11



Shutdown vs. Exit

- **Shutdown:**

A short-run decision not to produce anything because of
market conditions.

- A firm that shuts down temporarily must still pay its fixed costs.

- **Exit:**

A long-run decision to leave the market.

- A firm that exits the market does not have to pay any costs at all, fixed or variable.

12



A Firm's Short-run Decision to Shut Down

- If firm shuts down temporarily,
 - revenue falls by TR
 - costs fall by VC
- So, the firm should shut down if $TR < VC$.
- Divide both sides by Q : $TR/Q < VC/Q$
- So we can write the firm's decision as:

Shut down if $P < AVC$

13

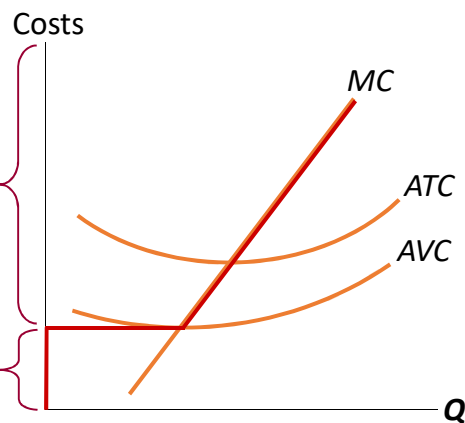


A Competitive Firm's SR Supply Curve

The firm's SR supply curve is the portion of its MC curve above

If $P > AVC$, then firm produces Q where $P = MC$.

If $P < AVC$, then firm shuts down (produces $Q = 0$).



14



The Irrelevance of Sunk Costs

- **Sunk cost:** a cost that has already been committed and cannot be recovered
- Sunk costs should be irrelevant to decisions; you must pay them regardless of your choice.
- *FC* is a sunk cost: The firm must pay its fixed costs whether it produces or shuts down.
- So, *FC* should not matter in the decision to shut down.



15



A Firm's Long-Run Decision to Exit

- If firm exits the market,
 - revenue falls by TR
 - costs fall by TC
- So, the firm should exit if $TR < TC$.
- Divide both sides by Q to rewrite the firm's decision as:

Exit if $P < ATC$



16



A New Firm's Decision to Enter Market

- In the long run, a new firm will enter the market if it is profitable to do so: if $TR > TC$.
- Divide both sides by Q to express the firm's entry decision as:

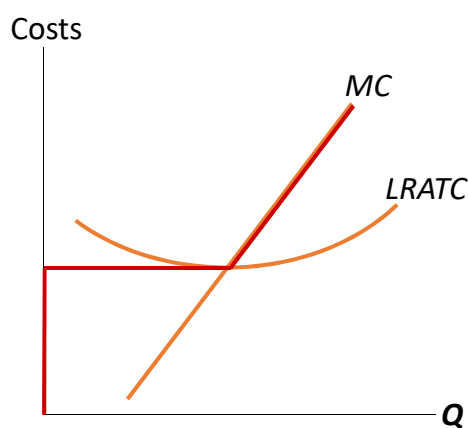
Enter if $P > ATC$

17



The Competitive Firm's Supply Curve

The firm's LR supply curve is the portion of its MC curve above $LRATC$.



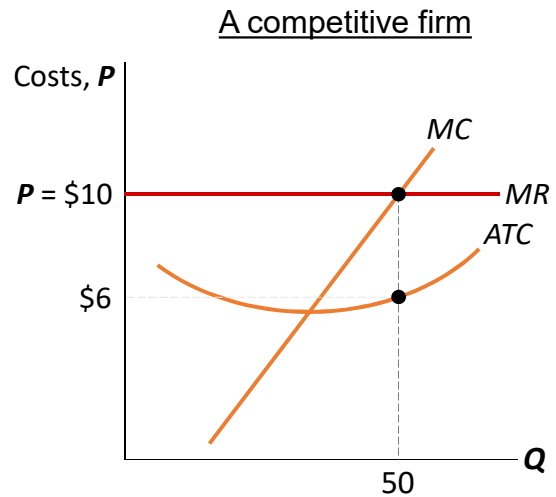
18



ACTIVE LEARNING 2A: Identifying a firm's profit

Determine
this firm's
total profit.

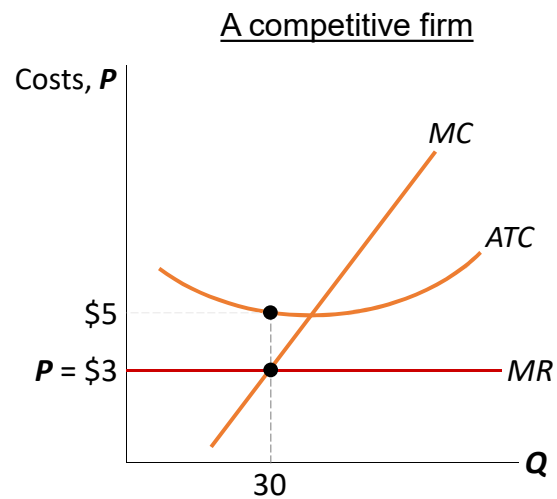
Identify the area on the
graph that represents
the firm's profit.



ACTIVE LEARNING 2B: Identifying a firm's loss

Determine
this firm's
total loss.

Identify the area on the
graph that represents
the firm's loss.





Market Supply: Assumptions

- 1) All existing firms and potential entrants have identical costs.
- 2) Each firm's costs do not change as other firms enter or exit the market.
- 3) The number of firms in the market is
 - fixed in the short run
(due to fixed costs)
 - variable in the long run
(due to free entry and exit)

21



The SR Market Supply Curve

- As long as $P \geq AVC$, each firm will produce its profit-maximizing quantity, where $MR = MC$.
- Recall from Unit 2:
At each price, the market quantity supplied is the sum of quantity supplied by each firm.

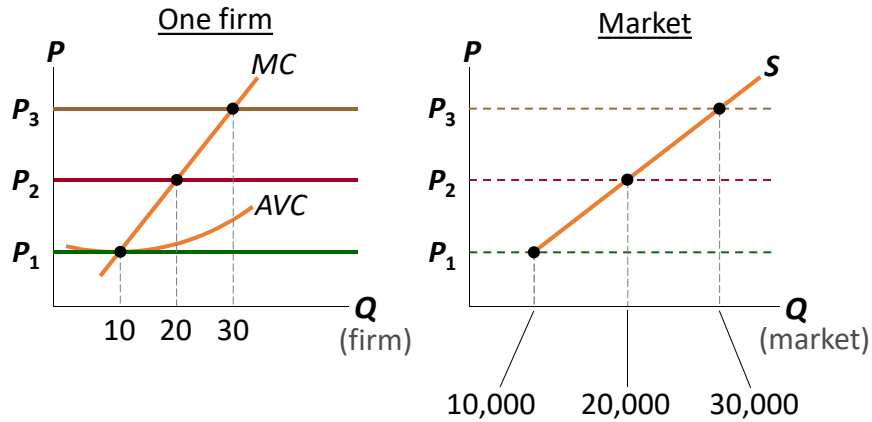
22



The SR Market Supply Curve

Example: 1000 identical firms.

At each P , market $Q^s = 1000 \times (\text{one firm's } Q^s)$



23



Entry & Exit in the Long Run

- In the LR, the number of firms can change due to entry & exit.
- If existing firms earn positive economic profit,
 - New firms enter.
 - SR market supply curve shifts right.
 - P falls, reducing firms' profits.
 - Entry stops when firms' economic profits have been driven to zero.

24



Entry & Exit in the Long Run

- In the LR, the number of firms can change due to entry & exit.
- If existing firms incur losses,
 - Some will exit the market.
 - SR market supply curve shifts left.
 - P rises, reducing remaining firms' losses.
 - Exit stops when firms' economic losses have been driven to zero.

25



The Zero-Profit Condition

- **Long-run equilibrium:**
The process of entry or exit is complete – remaining firms earn zero economic profit.
- Zero economic profit occurs when $P = ATC$.
- Since firms produce where $P = MR = MC$, the zero-profit condition is $P = MC = ATC$.
- Recall that MC intersects ATC at minimum ATC .
- Hence, in the long run, $P = \text{minimum } ATC$.

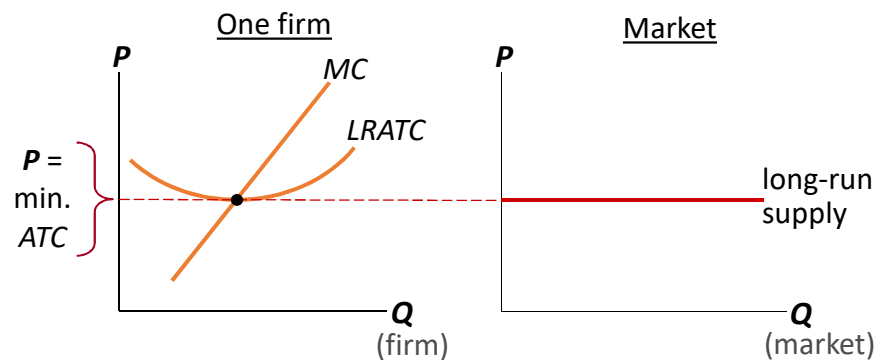
26



The LR Market Supply Curve

In the long run,
the typical firm
earns zero profit.

The LR market supply
curve is horizontal at
 $P = \text{minimum ATC}$.



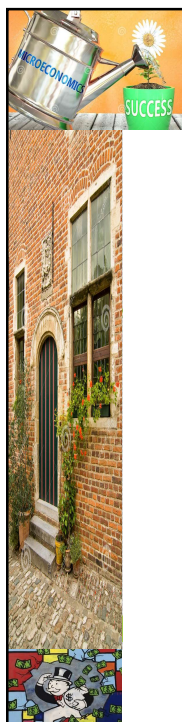
27



Why Do Firms Stay in Business if Profit = 0?

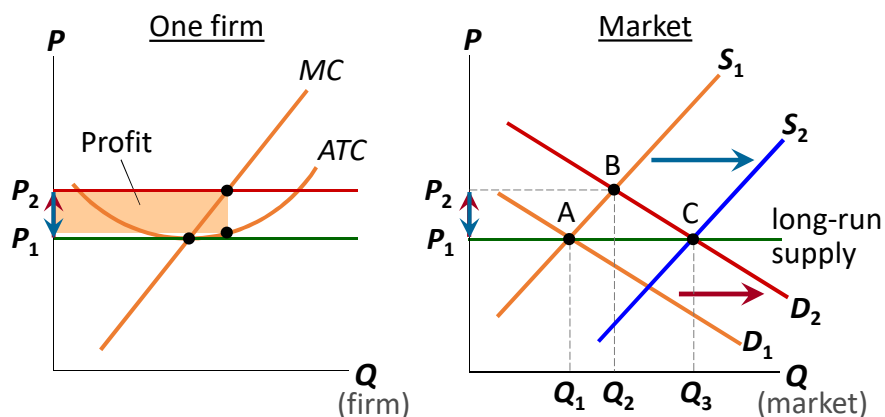
- Recall, economic profit is revenue minus all costs – including implicit costs, like the opportunity cost of the owner's time and money.
- In the zero-profit equilibrium, firms earn enough revenue to cover these costs.

28

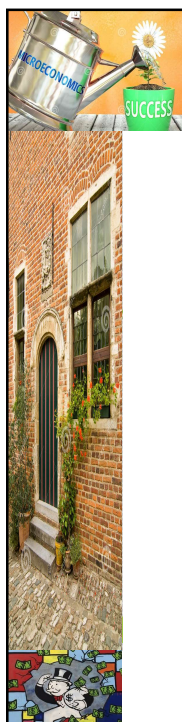


SR & LR Effects of an Increase in Demand

A firm begins in short-run equilibrium but then an increase in demand occurs, driving profits to zero. This leads to new entry, increasing the number of firms, which reduces price and restores long-run equilibrium.



29



Why the LR Supply Curve Might Slope Upward

- The LR market supply curve is horizontal if
 - all firms have identical costs, and
 - costs do not change as other firms enter or exit the market.
- If either of these assumptions is not true, then LR supply curve slopes upward.

30



1) Firms Have Different Costs

- As P rises, firms with lower costs enter the market before those with higher costs.
- Further increases in P make it worthwhile for higher-cost firms to enter the market, which increases market quantity supplied.
- Hence, LR market supply curve slopes upward.
- At any P ,
 - For the marginal firm,
 $P = \text{minimum } ATC$ and profit = 0.
 - For lower-cost firms, profit > 0 .

31



2) Costs Rise as Firms Enter the Market

- In some industries, the supply of a key input is limited (e.g., there's a fixed amount of land suitable for farming).
- The entry of new firms increases demand for this input, causing its price to rise.
- This increases all firms' costs.
- Hence, an increase in P is required to increase the market quantity supplied, so the supply curve is upward-sloping.

32



CONCLUSION: The Efficiency of a Competitive Market

- Profit-maximization: $MC = MR$
- Perfect competition: $P = MR$
- So, in the competitive eq'm: $P = MC$
- Recall, MC is cost of producing the marginal unit.
 P is value to buyers of the marginal unit.
- So, the competitive eq'm is efficient, maximizes total surplus.
- In the next chapter, monopoly: pricing & production decisions, deadweight loss, regulation.

33



SUMMARY

- For a firm in a perfectly competitive market, price = marginal revenue = average revenue.
- If $P > AVC$, a firm maximizes profit by producing the quantity where $MR = MC$. If $P < AVC$, a firm will shut down in the short run.
- If $P < ATC$, a firm will exit in the long run.
- In the short run, entry is not possible, and an increase in demand increases firms' profits.
- With free entry and exit, profits = 0 in the long run, and $P =$ minimum ATC .

34